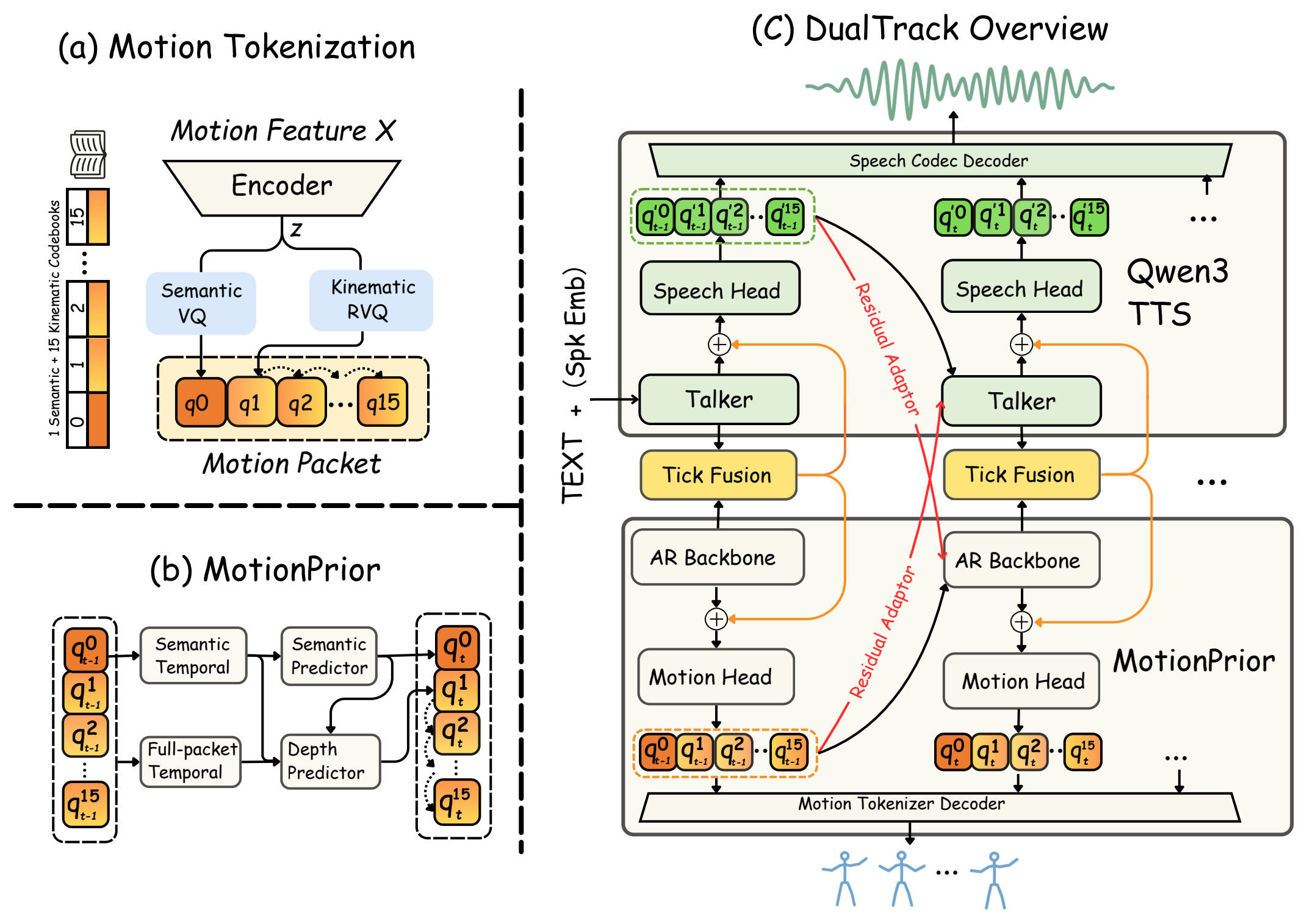

(a) Motion Tokenization
Motion Feature X
Encoder
z
Semantic VQ
Kinematic RVQ
1 Semantic + 15 Kinematic Codebooks
15
2
1
0
q0
q1
q2
q15
Motion Packet
(b) MotionPrior
Semantic Temporal
Semantic Predictor
Full-packet Temporal
Depth Predictor
(C) DualTrack Overview
Speech Codec Decoder
Speech Head
Speech Head
Qwen3 TTS
TEXT + (Spk Emb)
Talker
Talker
Residual Adaptor
Residual Adaptor
Tick Fusion
Tick Fusion
AR Backbone
AR Backbone
Motion Head
Motion Head
MotionPrior
Motion Tokenizer Decoder